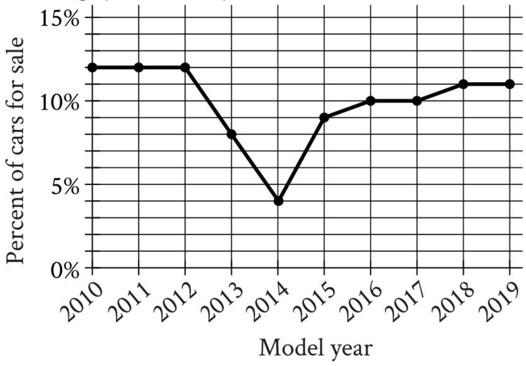
SAT 8

Math Module 2

Question #	ID
SAT8 M 2.1	4a2264b3

The line graph shows the percent of cars for sale at a used car lot on a given day by model year.



For what model year is the percent of cars for sale the smallest?

A. 2012

B. 2013

C. 2014

D. 2015

Question #	ID
SAT8 M 2.2	a08ma202

An object's speed is 64 **yards** per second. What is the object's speed, in <u>feet</u> per second?
$(1 \text{ yard} = 3 \text{ feet})$

(A) 61

(B) 67

(C) 94

(D) 192

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Question #	ID
SAT8 M 2.3	a08ma203

The lengths of two sides of a triangle are 4 centimeters and 6 centimeters. If the perimeter of the triangle is 18 centimeters, what is the length, in centimeters, of the third side of this triangle?

(A) 2

(B) 8

(C) 10

(D) 24

SAT8 M 2.4 70d9516e

A bus is traveling at a constant speed along a straight portion of road. The equation $d = 30t$ gives the distance d, in feet from a road marker, that the bus will be t seconds after passing the marker. How many feet from the marker will the bus be 2 seconds after passing the marker?

A. 30

B. 32

C. 60

D. 90

SAT8 M 2.5 c50ede6d

The total cost, in dollars, to rent a surfboard consists of a $\$25$ service fee and a $\$10$ per hour rental fee. A person rents a surfboard for t hours and intends to spend a maximum of $\$75$ to rent the surfboard. Which inequality represents this situation?

A. $10t \leq 75$

B. $10 + 25t \leq 75$

C. $25t \leq 75$

D. $25 + 10t \leq 75$

Question # **ID**

SAT8 M 2.6 a08ma206

	Live east of the river	Live west of the river	Total
Less than 40 years old	17	11	28
At least 40 years old	18	89	107
Total	35	100	135

The table summarizes members of a local organization by age and whether they live east or west of the river. If a member of the organization is selected at random, what is the probability that the selected member is at least 40 years old?

A) $\frac{28}{135}$

C) $\frac{100}{135}$

B) $\frac{35}{135}$

D) $\frac{107}{135}$

SAT8 M 2.7 aad7e1b9

The function f is defined by $f(x) = \frac{1}{10}x - 2$. What is the *y*-intercept of the graph of $y = f(x)$ in the *xy*-plane?

A. $(-2, 0)$

B. $(0, -2)$

C. $(0, \frac{1}{10})$

D. $(\frac{1}{10}, 0)$

SAT8 M 2.8 3ea87153

The function *g* is defined by $g(x) = x^2 + 9$. For which value of x is $g(x) = 25$?

A. 4

B. 5

C. 9

D. 13

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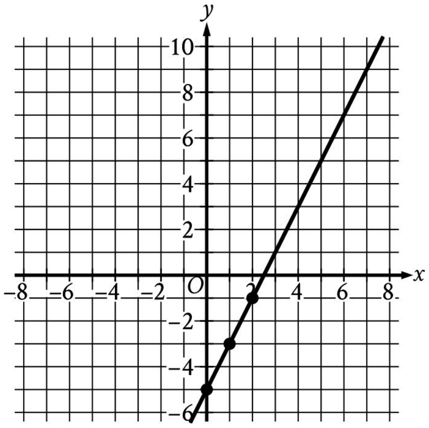
Question # **ID**

SAT8 M 2.9 4acd05cd



The graph shows the linear relationship between x and y. Which table gives three values of x and their corresponding values of y for this relationship?

A.

x	y
0	0
1	-7
2	-9

B.

x	y
0	0
1	-3
2	-1

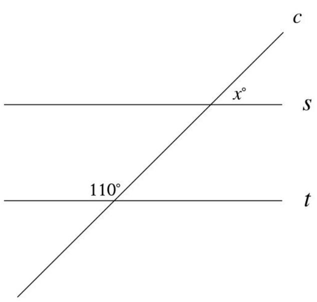
C.

x	y
0	-5
1	-7
2	-9

D.

x	y
0	-5
1	-3
2	-1

SAT8 M 2.10 cf0d3050



Note: Figure not drawn to scale.

In the figure shown, line c intersects parallel lines s and t. What is the value of x?

SAT8 M 2.11 7760c516

Each value in the data set shown represents the height, in centimeters, of a plant.

$$6, 10, 13, 2, 15, 22, 10, 4, 4, 4$$

What is the mean height, in centimeters, of these plants?

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Question #	ID
SAT8 M 2.12	a08ma212

$$f(x) = 4x + b$$

For the linear function f, b is a constant and $f(7) = 28$. What is the value of b?

(A) 0

(B) 1

(C) 4

(D) 7

SAT8 M 2.13 eac739b2

If $4x + 2 = 12$, what is the value of $16x + 8$?

A. 40

B. 48

C. 56

D. 60

SAT8 M 2.14 84e5e36c

$$y = 76$$
$$y = x^2 - 5$$

The graphs of the given equations in the *xy*-plane intersect at the point (x, y). What is a possible value of x?

A. $-\frac{76}{5}$

B. -9

C. 5

D. 76

SAT8 M 2.15 4ec95eab

$$y = -3x$$
$$4x + y = 15$$

The solution to the given system of equations is (x, y). What is the value of x?

A. 1

B. 5

C. 15

D. 45

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Question #	ID
SAT8 M 2.16	74c98c82

An event planner is planning a party. It costs the event planner a onetime fee of $\$35$ to rent the venue and $\$10.25$ per attendee. The event planner has a budget of $\$200$. What is the greatest number of attendees possible without exceeding the budget?

SAT8 M 2.17	717a1964

$$z^2 + 10z - 24 = 0$$

What is one of the solutions to the given equation?

SAT8 M 2.18	d8ace155

A company opens an account with an initial balance of $\$36{,}100.00$. The account earns interest, and no additional deposits or withdrawals are made. The account balance is given by an exponential function A, where $A(t)$ is the account balance, in dollars, t years after the account is opened. The account balance after 13 years is $\$68{,}071.93$. Which equation could define A?

A. $A(t) = 36{,}100.00(1.05)^t$

B. $A(t) = 31{,}971.93(1.05)^t$

C. $A(t) = 31{,}971.93(0.05)^t$

D. $A(t) = 36{,}100.00(0.05)^t$

SAT8 M 2.19	a4bd60a3

The perimeter of an equilateral triangle is 624 centimeters. The height of this triangle is $k\sqrt{3}$ centimeters, where k is a constant. What is the value of k?

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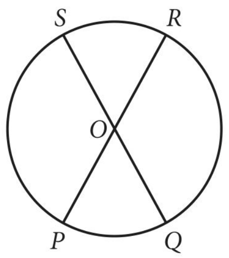
Question #	ID
SAT8 M 2.20	0815a5af



Note: Figure not drawn to scale.

The circle shown has center O, circumference 144π, and diameters $\overline{PR}$ and $\overline{QS}$. The length of arc PS is twice the length of arc PQ. What is the length of arc QR?

A. 24π

B. 48π

C. 72π

D. 96π

SAT8 M 2.21	b4a6ed81

The expression $90y^5 - 54y^4$ is equivalent to $ry^4(15y - 9)$, where r is a constant. What is the value of r?

SAT8 M 2.22	a08ma222

One leg of a right triangle has a length of 43.2 millimeters. The hypotenuse of the triangle has a length of 196.8 millimeters. What is the length of the other leg of the triangle, in millimeters?

A. 43.2

B. 120

C. 192

D. 201.5